

Line 15: Adds router middleware for extensibility.

Line 16: This is the fallback option when there is no content suitable from 'views'. This line enables static Web pages to be shown without extra lines of code.

Line 18: When no file is mentioned in the URL (after port), the page is directed to 'index.jade' in the views directory.

Lines 20-22: When an individual image file is requested (when the user clicks on a thumbnail), this sends the file to the browser.

Lines 24-26: Creates a server and waits for client requests.

Replace the contents of the *photoviewer/routes/index.js* file with the code given below:

```
1
2 /*
3  * GET home page.
4  */
5
6 var walk = require('walk');
7 var files = [];
8
9 // Walker options
10 var walker = walk.walk('./images/mypics',
11                       { followLinks: false });
12
13 walker.on('file', function (root, stat, next) {
14   files.push(root + '/' + stat.name);
15   next();
16 });
17
18 walker.on('end', function() {
19
20   exports.index = function(req, res){
21     res.render('index', { fileList: files });
22   };
23 };
```

Here is the explanation for the code given above.

Line 6: *walk* is a node package used to browse the file system.

Line 7: *files* is the array where we store the file names of a particular directory.

Line 10: A sub-directory called 'mypics' under the 'images' directory is selected to display pictures.

Lines 12 -17: Registers callback function for events on 'file' and 'end'. The 'file' event occurs for every file while traversing the directory. When we reach the end of directory listing, 'end' event is encountered.

Lines 20-22: The default page to be loaded is *index.jade*. We pass the list of files in the variable *filelist*. In the *index.jade* file, this variable will be used.

Replace the contents of the *photoviewer/views/index.jade* file with the code given below:

```
1 doctype 5
```

```
2 html
3 head
4   title Photo Viewer
5   link(rel='stylesheet',
6        href='/stylesheets/style.css')
7 body
8   h1(align="center") Photo Viewer
9   ul
10    each value in fileList.sort()
11      a(href= value): img(src= value width=80 height=50)
12  h6(align="center") (click on thumbnail to view
picture)
```

Here is the explanation for the code given above.

By default, Express creates two files '*index.jade*' and '*layout.jade*' in the views directory. For beginners who do not have a background in Jade, this could be confusing. The contents of the same two files have been merged to get the same result in the above case. HTML programmers will be able to understand what we plan to achieve from *index.jade*.

Lines 9-10: Jade has a looping structure just like other scripting languages. The *filelist* variable has come from *index.js*.

Enhancing the photoviewer project

To enhance the photoviewer project, it is recommended that you learn the Jade Template language. Jade is easy to code due to its shorthand notation and the powerful embedding of programming logic.

In addition, you can use the Express form handling function. In this project, we have only used the HTTP request type GET. To perform form handling you can use the POST or PUT operations. To keep the article concise, I have not covered the users' section of Express, which enables websites like blogs to be developed. Express also has features such as *fileupload*, which will enhance your photoviewer functionality to upload pictures from the browser to the server. Routes is another powerful feature, though a little complex to grasp at the first go. 

References

- [1] Express home-<http://expressjs.com>
- [2] Excellent video tutorial to get started on Express <http://www.youtube.com/watch?v=eD2l0zAJM5g>
- [3] Tutorial on Express.js explaining connect and routes-<http://evanhahn.com/understanding-express-js/>
- [4] Introduction to middleware-http://stephensugden.com/middleware_guide/

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